Euler Equation in the Optimal Growth Model

(Dynamic Programming Derivation)

Planner's Problem (Deterministic)

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t, \quad k_0 \text{ given}$$

Bellman Equation

Define the value function

$$V(k) = \max_{k' \geq 0} \left\{ u\big(f(k) + (1-\delta)k - k'\big) + \beta V(k') \right\}$$

Let consumption be

$$c = f(k) + (1-\delta)k - k'.$$

First-Order Condition (interior)

Differentiate with respect to k':

$$-u'(c) + \beta V'(k') = 0$$

$$u'(c_t) = \beta V'(k_{t+1})$$

Envelope Condition

Differentiate the Bellman equation with respect to k:

$$V'(k) = u'(c)\left[f'(k) + (1-\delta)\right]$$

Euler Equation

Substitute the envelope condition forward one period:

$$V'(k_{t+1}) = u'(c_{t+1})\left[f'(k_{t+1}) + (1-\delta)\right]$$

Therefore,

$$\boxed{u'(c_t) = \beta u'(c_{t+1}) \left[f'(k_{t+1}) + (1 - \delta) \right]}$$

Economic Interpretation

The marginal utility cost of consuming one unit today equals the discounted marginal utility benefit of investing it and consuming the return tomorrow.

CRRA Utility and Consumption Growth

Assume constant relative risk aversion (CRRA) preferences:

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \qquad \sigma > 0,\ \sigma \neq 1.$$

Then the marginal utility is

$$u'(c) = c^{-\sigma}.$$

Substituting $u'(c) = c^{-\sigma}$ into the Euler equation

$$u'(c_t) = \beta u'(c_{t+1}) \left[f'(k_{t+1}) + (1 - \delta) \right],$$

yields

$$c_t^{-\sigma} = \beta \, c_{t+1}^{-\sigma} \left[f'(k_{t+1}) + (1 - \delta) \right].$$

Rearranging gives the consumption growth form:

$$\left(\frac{c_{t+1}}{c_t} \right)^{\sigma} = \beta \left[f'(k_{t+1}) + (1 - \delta) \right],$$

and hence

$$\boxed{\frac{c_{t+1}}{c_t} = \left(\beta \left[f'(k_{t+1}) + (1 - \delta) \right] \right)^{1/\sigma}}.$$

Steady State

In a steady state, $c_{t+1} = c_t$, so the Euler equation implies

$$1 = \beta \left[f'(k^*) + (1 - \delta) \right],$$

or equivalently

$$\boxed{f'(k^*) = \frac{1}{\beta} - (1 - \delta)}.$$

Log Utility Case ($\sigma = 1$)
If $\sigma = 1$, then $u(c) = \ln c$ and $u'(c) = 1/c$, so

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \left[f'(k_{t+1}) + (1 - \delta) \right] \quad \Rightarrow \quad \boxed{\frac{c_{t+1}}{c_t} = \beta \left[f'(k_{t+1}) + (1 - \delta) \right]}.$$